

Stephen Boileau

Technical Artist & Developer | Unity / Unreal Engine | Pipeline Creation | Real-Time Optimization

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PROFESSIONAL SUMMARY

Technical Artist and Developer with 10+ years of experience building real-time, interactive, and pre-rendered experiences for themed entertainment, robotics, AR/VR, and mobile products. Strong background in Unity C#, Unreal Engine C++/Blueprints, Maya/Python tooling, repository management, asset export/import workflows, pipeline creation, UI/UX implementation, and real-time performance optimization. Known for bridging artists, designers, engineers, and creative directors while building scalable, artist-friendly workflows for production teams. Professional proficiency in English.

Core Qualifications

- 10+ years as a Technical Artist / Developer across themed entertainment, robotics, AR/VR, mobile, and real-time production.
- 8+ years of Unity game engine experience with C# gameplay, UI, AR/VR, mobile deployment, and cross-platform integration.
- 2D graphics software: Photoshop, Illustrator, and Adobe Creative Suite for UI/support assets, production prep, and creative lab workflows.
- 3D graphics software: Autodesk Maya for rigging, skinning, modeling support, UVs, animation preparation, validation, and export.
- Repository management: Perforce, Git, and Plastic SCM, including source-control setup, branching/workflow support, and team coordination.
- Asset exporting/importing, optimization, and pipeline creation for Maya-to-Unity and Maya-to-Unreal workflows, real-time characters, VR, and client review media.
- Strong understanding of real-time performance constraints: mobile, VR/Meta Quest, Android/iOS, real-time robotics, asset budgets, runtime stability, and optimization techniques.
- UI/UX implementation experience: Unity UI, mobile controls, VR/client-facing review tools, in-engine interfaces, and artist-friendly tool design.
- AI-driven tools/workflows: hands-on curiosity and active exploration of AI-assisted content creation, reference generation, voice/image workflows, and pipeline-efficiency opportunities.

PROFESSIONAL EXPERIENCE

Super 78 — Los Angeles, CA | Technical Artist & Developer

December 2021 – Present

- Delivered immersive themed entertainment experiences using Unity3D C# and Unreal Engine C++/Blueprints, supporting both real-time and pre-rendered production needs.
- Created and maintained asset pipeline workflows for Maya-to-Unity and Maya-to-Unreal production, including asset preparation, validation, export/import, naming standards, and directory structures.
- Built Python-based batch rendering and file-processing tools that standardized outputs, reduced manual artist intervention, and improved review-media consistency.
- Designed and maintained Unreal Engine rendering and capture pipelines for high-resolution output, client review, presentations, and marketing delivery.
- Implemented large-resolution rendering workflows using OWL (Off World Live) for Unreal Engine to support high-quality capture and nonstandard output requirements.
- Integrated Unreal Engine MetaHuman workflows and real-time character pipelines, supporting high-fidelity characters in live, rendered, and review outputs.
- Developed and optimized cross-platform VR applications for Oculus / Meta Quest, with attention to frame rate, device constraints, asset budgets, runtime stability, and performance bottlenecks.
- Implemented UI/UX-facing workflows for internal tools, client-facing VR review apps, and artist-friendly production tools, prioritizing simple controls for non-technical users.
- Established and maintained source-control workflows using Perforce, Git, and Plastic SCM, supporting repository management for multi-discipline production teams.
- Maintained, refactored, and extended legacy projects while supporting new feature development under tight production schedules.

Embodied Inc. — Pasadena, CA | Technical Artist

June 2018 – December 2021

- Prepared and integrated character rigs and animation assets for real-time robotic systems, supporting the production and ongoing development of Moxie.
- Developed Python automation tools to streamline asset processing, validation, export consistency, and production handoff between animation, art, and engineering.
- Collaborated through Git-based workflows with engineering, animation, and design teams to align technical solutions with creative and functional requirements.
- Improved iteration speed and asset stability through tooling, documentation, workflow optimization, and repeatable pipeline practices.
- Supported real-time character performance constraints where animation quality, runtime reliability, and production efficiency were all critical.

DeepBlue Worlds — Los Angeles, CA | Junior Technical Artist

February 2014 – June 2018

- Built interactive Unity C# features and supported Autodesk Maya asset workflows for mobile, AR, and real-time projects.
- Worked on consumer-facing products including Dino on My Desk, contributing gameplay integration, UI implementation, asset preparation, and Android/iOS deployment support.

- Optimized assets and real-time content for mobile performance constraints, including efficient export/import practices and engine-ready content preparation.
- Gained hands-on production experience across art, code, UI, rigging, animation support, and real-time engine workflows.

Mt. San Antonio College — Walnut, CA | Lab Technician / Teacher Assistant

August 2013 – December 2016

- Assisted professors teaching Autodesk Maya, introductory modeling, and game-development fundamentals.
- Supported students in creative labs using Maya, Photoshop, Illustrator, and Adobe Creative Suite.
- Provided technical guidance on 3D workflows, real-time fundamentals, asset preparation, and production problem solving.

SELECTED PROJECT HIGHLIGHTS

- VR Theater Visualizer: built a scaled Maya-to-Quest review workflow with curved-screen UV layouts, media playback, and a simple file-picking UX so non-technical clients could review content in headset.
- Real-Time Character System: supported live Unreal/MetaHuman-style character workflows with technical art, animation, and runtime reliability requirements.
- Interactive Cruise Dining Experience: supported interactive media workflows using TouchDesigner/OpenCV-style computer vision concepts to drive guest-responsive table graphics.
- Dino on My Desk: shipped mobile AR experience as a Unity developer / technical artist, including gameplay integration, UI, real-time assets, and Android/iOS delivery.

TECHNICAL SKILLS

Engines: Unity3D / Unity 6, C#, Unreal Engine, Blueprints, C++, real-time rendering, AR/VR, mobile deployment

Pipeline & Tools: Pipeline creation, Python, PyMEL, Maya tools, batch rendering, automated file processing, asset validation, export/import workflows

Optimization: Real-time performance constraints, mobile/VR optimization, Meta Quest constraints, asset budgets, LODs, runtime stability, rendering/capture performance

2D / 3D Software: Autodesk Maya, Photoshop, Illustrator, Adobe Creative Suite, rigging, skinning, modeling support, UV mapping

Repository Management: Perforce, Git, Plastic SCM, branching workflows, production source control, legacy-project maintenance

UI / UX Implementation: Unity UI, mobile controls, VR interfaces, in-engine tool UX, client-facing review workflows, artist-friendly tools

AI-Assisted Workflows: AI-driven content creation curiosity, reference generation, image/voice workflow exploration, automation opportunities for pipeline efficiency

Language: English - professional proficiency